

Introduction

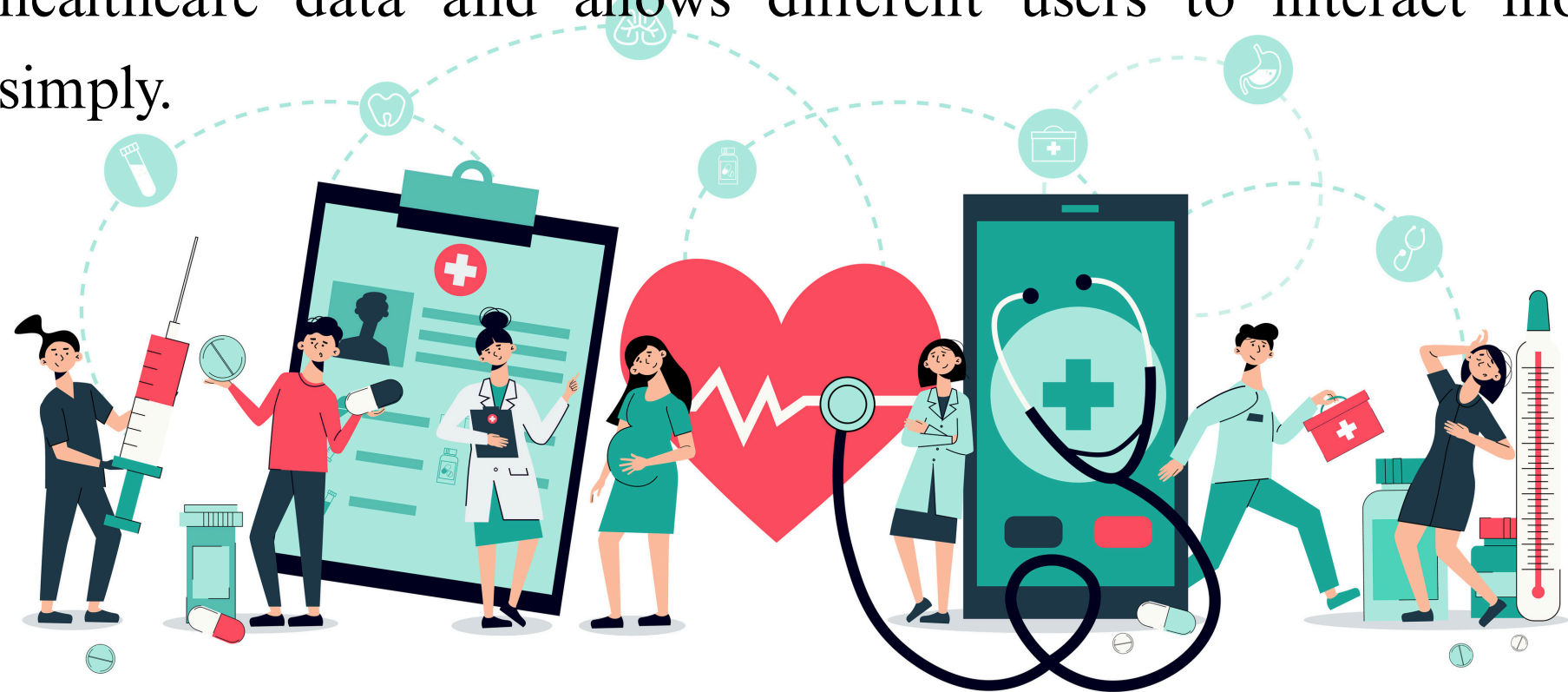
There is often a lack of order and connection between doctors, patients, labs, pharmacies and insurers in the healthcare industry. When things are not addressed efficiently, treatment may not start on time, the results may not be accurate and the costs for administration may go up. The purpose of this platform is to facilitate communication and combine different medical services while allowing all involved parties to easily access medical records.

Problem Statement

Across healthcare systems in developing countries, records related to patients are often held on unconnected platforms such as hospitals, laboratories, pharmacies and insurance companies. This brings about important problems and consequences.

- Due to not having all the patient’s information right away, healthcare providers can miss optimal times to diagnose or treat patients.
- Lack of Integration: Without integrated technologies connecting all staff, there is little opportunity for smooth teamwork, resulting in more redundant procedures and added paperwork.
- Incorrect data or inaccuracies may occur when care providers have to manually record information, leading to problems for patients.
- Most patients are unable to view all their health records which lowers their chances of being actively involved in their health decisions.

As a result, we need a unified digital platform that gathers all healthcare data and allows different users to interact more simply.



Centralized Healthcare Eco-system

A Centralized Health Platform for Integrated Care:  
Enhancing Stakeholder Coordination and Data  
Interoperability in Healthcare Ecosystems

Justification

The healthcare industry across the globe is shifting toward a digital approach. Much of Sri Lanka’s system relies on paper records, separated clinics and manual checks between providers.

This project includes:

- Decreased mistakes when writing prescriptions
- Easier and quicker insurance approvals
- Increased involvement of patients
- Use electronic methods to order laboratory tests and medicine.

Objectives

To ensure a simple, protected and central platform for digital health that can be accessed by all.

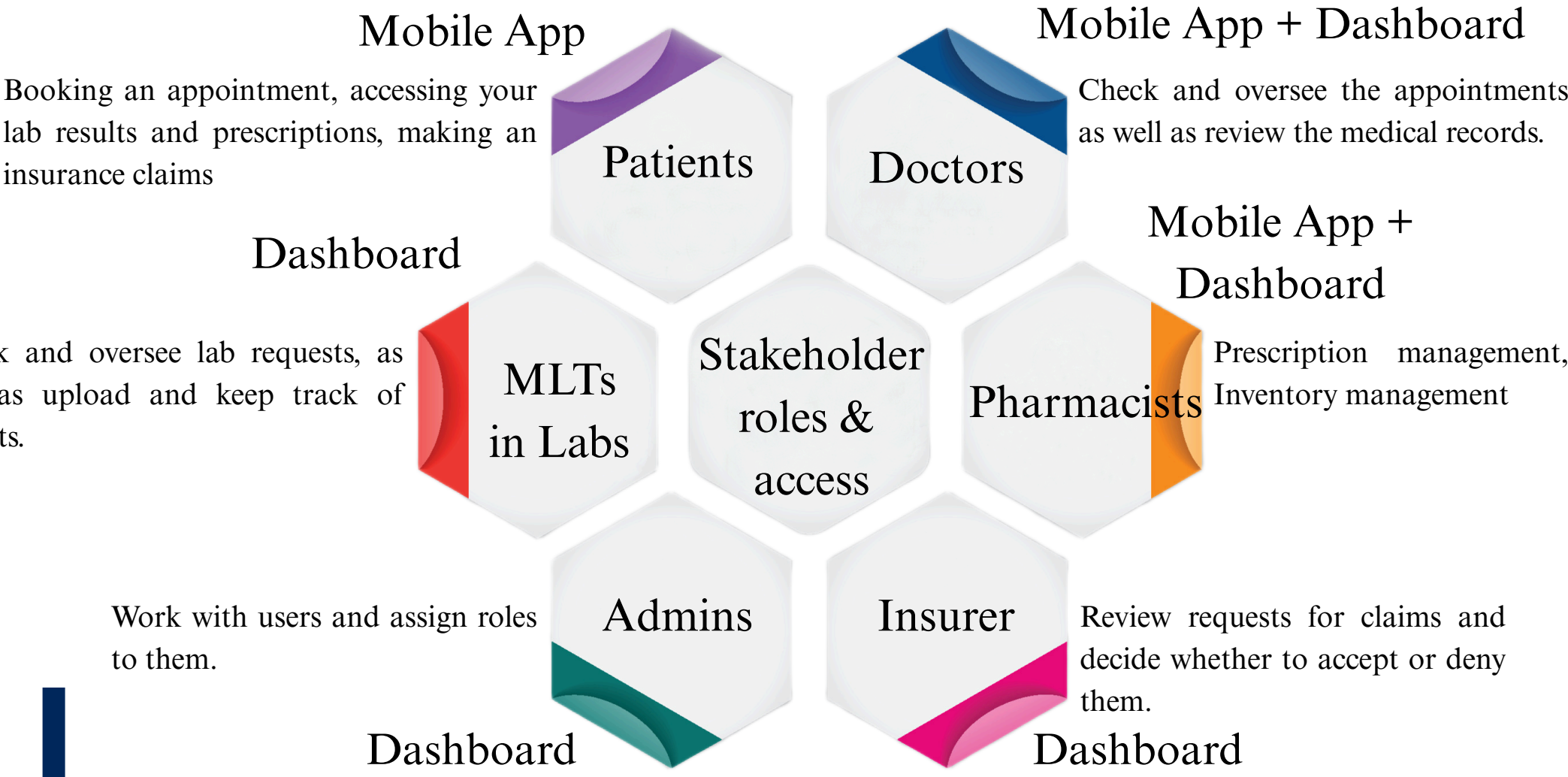
- Combines hospitals, laboratories, pharmacies and insurance providers into a single network.
- Allows Authorized users to view the most current and thorough records of each patient virtually any time.
- Makes It Easier to Communicate: Helps healthcare professionals and patients communicate with each other without hassle.
- Makes Information Available to Patients: Helps people access their health information which aids them in decision-making.

The purpose of these objectives is to support better service in healthcare, lessen mistakes and help patients enjoy better care.

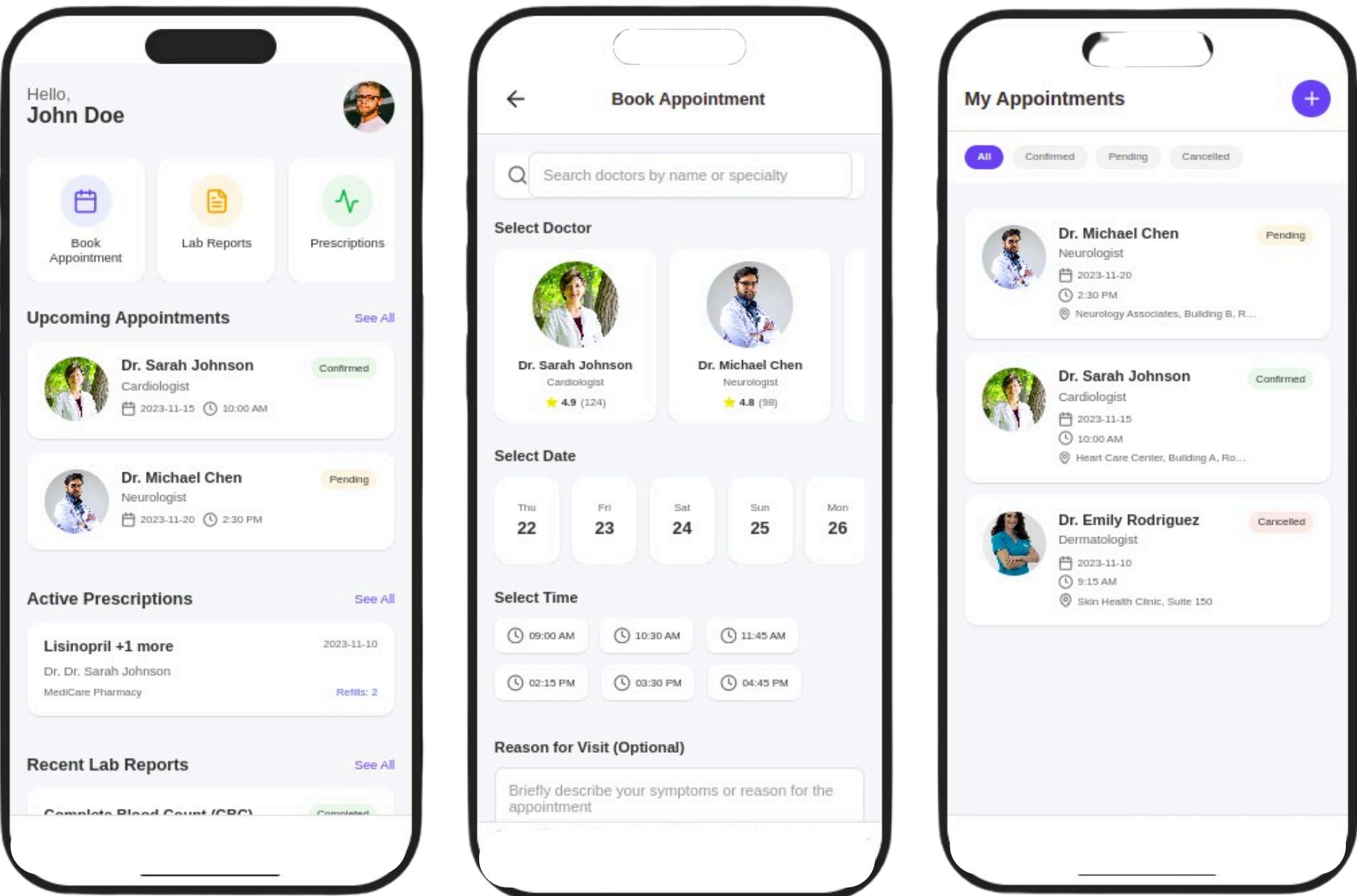
Methodology

System Architecture:

- Frontend:  
Mobile app: React Native Expo  
Dashboard: Next.js
- Backend: Node.js + Express.js
- Database: PostgreSQL
- Authentication: JWT Tokens



Results



Conclusion

This project is developed to provide a central unified platform where patients, doctors, pharmacists, lab technicians, insurers and administrators all connect securely. It deals with main challenges in healthcare by facilitating effective communication and sharing of important information.

Experts believe that applying this platform will bring better outcomes for patients, less paperwork for offices and connect all aspects of healthcare.

References

• <https://www.who.int/docs/default-source/documents/gs4dhdaa2a9f352b0445bafbc79ca799dce4d.pdf>  
• [https://ecqi.healthit.gov/fhir?qt-tabs\\_fhir=about](https://ecqi.healthit.gov/fhir?qt-tabs_fhir=about)  
• <https://www.health.gov.lk/wp-content/uploads/2022/10/English-Perfomence-Progress-Report-compressed.pdf>  
• <https://nextjs.org/docs>  
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